

**Diagnostic accuracy of pelvic imaging for acute pelvic inflammatory disease:
protocol for a systematic review and meta-analysis**

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INTRODUCTION

Rationale

Pelvic inflammatory disease (PID) is an infection of the female upper reproductive tract (i.e., the endometrium, fallopian tubes, ovaries, or pelvic peritoneum). In particular, acute (≤ 30 days duration) PID is common among sexually active young and adolescent women^{1 2}, and presents with fever, lower abdominal pain, and abdominal vaginal discharge^{3 4}. Acute PID is one of the most common causes of the acute abdomen at an emergency department^{5 6}, and not only gynecologists but also emergency or general physicians often need to diagnose and treat this condition⁷. In addition, PID has increased the risk of infertility⁸, ectopic pregnancy⁹, and chronic pelvic pain¹⁰, and delayed care of PID has been associated with worse long-term outcomes¹¹. Thus, major international guidelines recommend that treatment for PID should not be delayed^{3 4}.

Women of reproductive age who present acute lower abdominal pain with a clinical suspicion of acute PID need gynecological examination (e.g., pelvic examination or speculum examination) for the clinical diagnosis of PID according to the Centers for Disease Control and Prevention guideline issued in 2015³. Instead, imaging modalities such as computed tomography (CT) and magnetic resonance imaging (MRI) can investigate causes of abdominal pain which are not only non-gynecological causes but also acute PID. If these findings suggest acute PID, physicians may administer antibiotics for this condition without gynecological examination.

It is not well-known how accurate imaging modalities are for the diagnosis of acute PID. If these modalities have enough diagnostic accuracy, they can alternate diagnostic tools from the gynecological examination. This review will investigate the diagnostic accuracy of CT and MRI, and determine whether these imaging modalities are

replaceable for gynecological examinations.

Clinical role of the index test

Pelvic imaging such as CT and MRI can help to identify many causes of the acute abdomen including acute PID^{12 13 14 15}. For example, the common findings of acute PID include thickening of the uterosacral ligaments or fallopian tube, pelvic fat stranding with obscuration of fascial planes, reactive lymphadenopathy, and pelvic free fluid with an enhanced-contrast CT¹⁶. In general, young female patients with acute lower abdominal pain are consulted for a gynecologist to undergo pelvic examination and speculum examination if they are suspected of PID. Additional evaluation using transvaginal ultrasound may be performed to identify a more accurate diagnosis. However, consultation for gynecologists is not always available in limited settings (e.g., in rural areas or physician offices)^{17 18}. In addition, the clinical manifestation of acute PID can mimic other causes of acute abdomen occasionally requiring surgical intervention (e.g., acute appendicitis, diverticulitis, and ovarian cyst rupture/torsion)¹⁹²⁰. If imaging modalities such as CT and MRI accurately identify acute PID without gynecological examination, they are likely to become alternative diagnostic tools. If so, emergency or general physicians will treat this condition early without gynecological examination.

OBJECTIVE

This systematic review and meta-analysis aims to evaluate the diagnostic accuracy of pelvic imaging for acute PID among women of reproductive age.

METHODS

This systematic review was adhered to the Preferred Reporting Items for a Systematic Review and Meta-analysis of Diagnostic Test Accuracy Studies (PRISMA-DTA) statement²¹.

Criteria for considering this review

Types of studies

We will include all studies on the diagnostic accuracy of pelvic imaging for evaluating acute PID. We will include the studies designed as follows: prospective or retrospective observational (cohort or cross-sectional) studies, secondary analysis of randomized controlled trials (RCTs), and diagnostic case-control studies. We will not exclude studies based on language, countries, year of publication, or publication status. We will exclude studies for which the true positive, true negative, false positive, and false negative cannot be calculated.

Participants

The target participants are women of reproductive age with acute lower abdominal pain who are suspicious of PID and required gynecological examinations. We will exclude pregnant patients.

Target condition

The target condition is acute PID including any combination of endometritis, salpingitis, tubo-ovarian abscess, and pelvic peritonitis. We will exclude studies that explicitly focused on the diagnosis of perihepatitis (i.e., Fitz-Hugh-Curtis syndrome) because of

differences in clinical manifestation. We will define ‘acute’ as symptoms lasting 30 days or less¹³.

Setting

We expect studies conducted in emergency settings, however, we will accept any type of settings.

Index tests

The index test assessed in this review is pelvic imaging for acute PID. We will assess pelvic imaging modalities as contrast-enhanced CT and MRI. Based on the previous literature and national guidelines³⁻¹⁶, the positive findings are defined as follows: regarding an enhanced-contrast CT, thickening of the uterosacral ligaments or fallopian tube, pelvic fat stranding with obscuration of fascial planes, and pelvic free fluid; regarding MRI, thickened, swollen, or fluid-filled tubes with or without free pelvic fluid or tubo-ovarian complex. We will also include other positive findings of each imaging modality as the authors defined. We will allow the judgment of these positive findings by any physicians in included studies.

Reference standard

The reference standard is gynecological examinations defined as examinations by gynecologists using standard diagnostic criteria³ with or without laparoscopy or transcervical endometrial biopsy. We also consider other diagnostic criteria for acute PID as authors defined (e.g., the microbiologic basis of diagnosis²²).

Search methods for identification of studies

Electronic searches

We will conduct a literature search to identify all published and unpublished eligible studies. We will translate non-English articles and thoroughly assess them for potential inclusion as necessary. We will search in the following electronic databases: MEDLINE, EMBASE, and Cochrane Central Register of Controlled Trials. We will also search for an international clinical trials registry platform (ICTRP) and ClinicalTrials.gov to find ongoing and unpublished studies. We will check the reference lists of all included studies, clinical practice guidelines (e.g., Centers for Disease Control and Prevention Sexually Transmitted Disease Treatment Guideline³ and European Guideline⁴), and relevant systematic reviews to identify additional studies missed during the original electronic searches.

Data collection and analysis

Selection of studies

Two pairs of three review authors (YO, KY, and TA) will independently screen the study eligibility. The authors will screen the results of the search according to our search strategy by a two-step process with initial title and abstract screening followed by full-text screening. If the authors disagree, they will discuss to resolve the disagreement or consult with a third-party reviewer. We will record the reasons for the exclusion of ineligible full text studies. We will exclude all duplicate publications. We will record the selection process in sufficient detail to complete the PRISMA flow diagram and the characteristics of the excluded studies table.

Data extraction and management

Two pairs of three review authors (YO, KY, and TA) will develop the data extraction sheet including the following information.

1. Characteristic of studies: authors, year of publication, country, study design, sample size, clinical setting, number of studies, funding source, and journal citation
2. Participants: inclusion and exclusion criteria, the total number of study participants, number of drop-outs with reason, and characteristics of participants (e.g., gender, and age)
3. Index test: timing of sampling, methods of examination, time to results, and who performed or judged it
4. Reference standard: timing of sampling, methods of examination, time to results, and who performed or judged it
5. Outcomes: from the 2×2 table, we will calculate true positives, false positives, true negatives, and false negatives

Assessment of methodological quality

Two pairs of three review authors (YO, KY, and TA) will independently assess the risk of bias and applicability, and report the results according to the QUADAS-2 tool²³. We will revise the signaling questions in the QUADAS-2 according to this review question. We will count the number of ongoing and unpublished studies for the assessment of publication bias.

Statistical analysis and data synthesis

We will present the included available studies summarized in two tables. The first table

will summarize the study designs, participants, index test details, sample types, and the reference standards. The second one will summarize the details on the quality of studies evaluated by QUADAS-2.

We will analyze an enhanced-contrast CT and MRI, together. We will compare the results of pelvic imaging to that of the reference standard. Data for 2×2 tables of pelvic imaging against the reference standard will be extracted from each study. We will present a summary table of evidence and estimates of sensitivity and specificity with 95% confidence intervals of pelvic imaging per study using forest plots to inspect between-study variability. We will also present estimates of the accuracy of each study using a scatter plot in a receiver operating characteristic (ROC) to visually assess the correlation between both indices.

We will conduct a meta-analysis for an enhanced-contrast CT and MRI, together. In the meta-analysis, we will use a hierarchical summary ROC (HSROC) model to report the summary ROC and the point estimate of sensitivity and specificity with 95% credible interval (CrI). We will fix the value of sensitivity to calculate the point estimate of specificity.

All analyses will be performed using SAS Universal Edition, Review Manager 5.4.1 (Cochrane Collaboration, London, United Kingdom), and the calculator for the summary points from the HSROC model (CAST-HSROC) to report the point estimate of either sensitivity or specificity with 95% CrI for a fixed specificity or sensitivity value on the HSROC model²⁴.

Investigations of heterogeneity

Whenever possible, we will conduct subgroup analyses on the following groups.

1. Type of participants (outpatients versus inpatients) ³
2. Type of reference standard (non-invasive procedures (e.g., clinical diagnostic criteria) versus invasive procedure (i.e., laparoscopy or transcervical endometrial biopsy))
3. Type of pelvic imaging (contrast-enhanced CT versus MRI).

We will report the results along with an enhanced-contrast CT and MRI regarding the type of participants and type of reference standard. We will perform a meta-regression to investigate the relative diagnostic odds ratio and will report the odds ratio with 95% confidence interval.

Sensitive analysis

We will assess the robustness to explore diagnostic accuracy for pelvic imaging by excluding case-control studies and studies judged to be a high risk of bias.

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Appendix 1. MEDLINE via PubMed search strategy

#1 Tomography, X-ray Computed[mh]
#2 Magnetic Resonance Imaging[mh]
#3 "computed tomography"[tiab]
#4 "magnetic resonance imaging"[tiab]
#5 CT[tiab]
#6 MRI[tiab]
#7 Pelvic Inflammatory Disease[mh]
#8 "pelvic inflammatory disease"[tiab]
#9 PID[tiab]
#10 "tubo-ovarian abscess"[tiab]
#11 "pelvic peritonitis"[tiab]
#12 endometritis[tiab]
#13 salpingitis[tiab]
#14 #1 OR #2 OR #3 OR #4 OR #5 OR #6
#15 #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13
#16 #14 AND #15

Appendix 2. EMBASE via Dialog search strategy

S1 EMB.EXACT.EXPLODE("x-ray computed tomography")
S2 EMB.EXACT.EXPLODE("nuclear magnetic resonance imaging")
S3 AB("computed tomography") OR TI("computed tomography")
S4 AB("magnetic resonance imaging") OR TI("magnetic resonance imaging")
S5 AB(CT) OR TI(CT)
S6 AB(MRI) OR TI(MRI)
S7 EMB.EXACT.EXPLODE("pelvic inflammatory disease")
S8 AB("pelvic inflammatory disease") OR TI("pelvic inflammatory disease")
S9 AB(PID) OR TI(PID)
S10 AB("tubo-ovarian abscess") OR TI("tubo-ovarian abscess")
S11 AB("pelvic peritonitis") OR TI("pelvic peritonitis")
S12 AB(endometritis) OR TI(endometritis)
S13 AB(salpingitis) OR TI(salpingitis)
S14 S1 OR S2 OR S3 OR S4 OR S5 OR S6
S15 S7 OR S8 OR S9 OR S10 OR S11 OR S12 OR S13
S16 S14 AND S15

Appendix 3. Cochrane Central Register of Controlled Trials (CENTRAL) search strategy

([mh Tomography, X-ray Computed] OR [mh Magnetic Resonance Imaging] OR "computed tomography":ti,ab,kw OR "magnetic resonance imaging":ti,ab,kw OR CT:ti,ab,kw OR MRI:ti,ab,kw) AND ([mh Pelvic Inflammatory Disease] OR "pelvic inflammatory disease":ti,ab,kw OR PID:ti,ab,kw OR "tubo-ovarian abscess":ti,ab,kw OR "pelvic peritonitis":ti,ab,kw OR endometritis:ti,ab,kw OR salpingitis:ti,ab,kw)

Appendix 4. World Health Organization International Clinical Trials Registry Platform (WHO ICTRP) search strategy

("pelvic inflammatory disease" OR "endometritis" OR "salpingitis" OR "tubo-ovarian abscess" OR "pelvic peritonitis")

AND

("computed tomography" OR "magnetic resonance imaging")

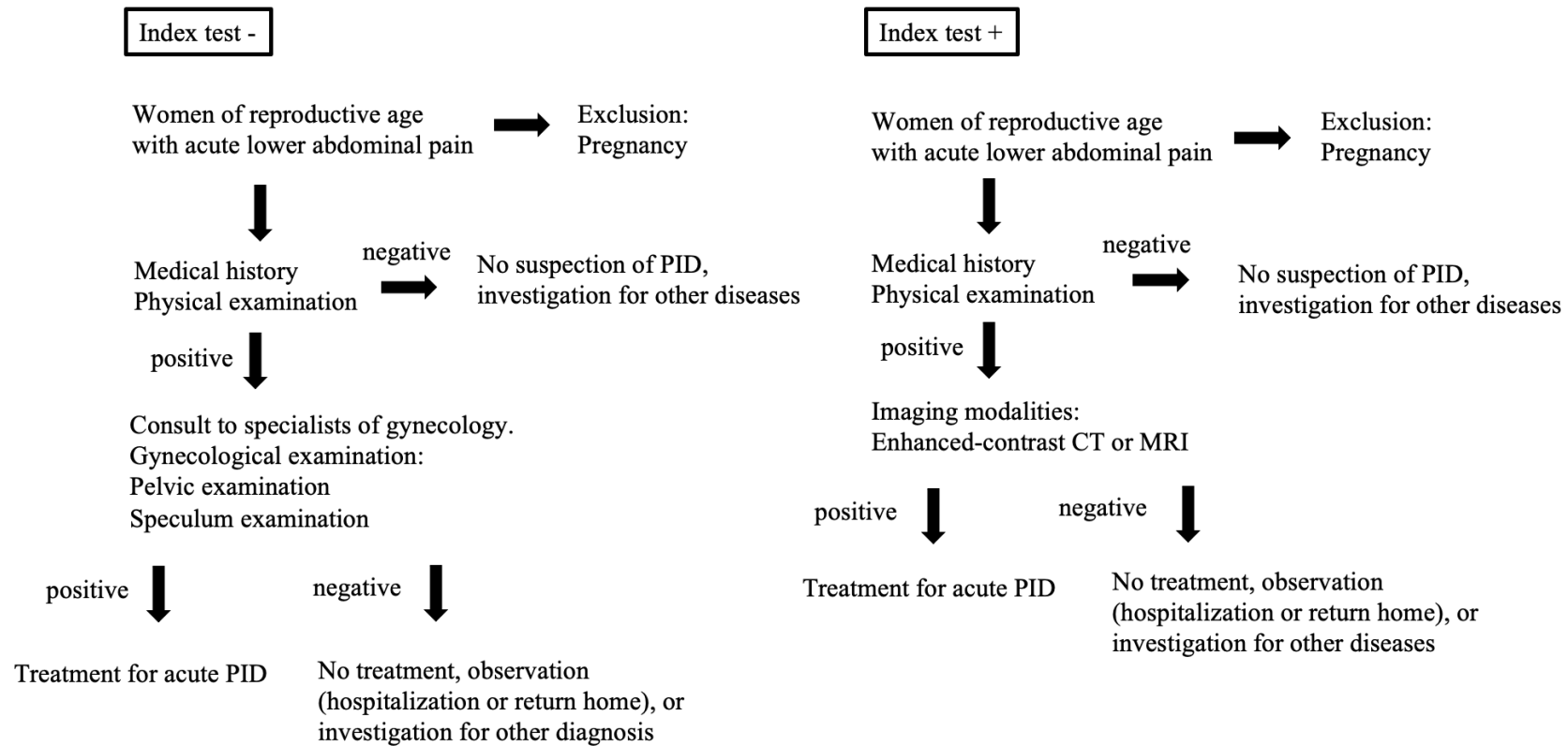
Appendix 5. ClinicalTrials.gov search strategy

Condition or disease: pelvic inflammatory disease OR endometritis OR salpingitis OR tubo-ovarian abscess OR pelvic peritonitis

Intervention: computed tomography OR magnetic resonance imaging

Appendix 6. Clinical pathway for this review question

Setting: emergency settings



Abbreviation: PID ; pelvic inflammatory disease, CT ; computed tomography, MRI ; magnetic resonance imaging